AI CAPSTONE PROJECT

Exploring APIs with Python – Fetching Real-Time Data

Student Name: Cynthia Dalmas

School: Moringa School

Course: AI Essentials for Software Development

PROJECT INSTRUCTIONS

1. Title & Objective

- Example: “Getting Started with TailwindCSS in React – A Beginner’s Guide”
- What technology did you choose?
- Why did you choose it?
- What’s the end goal (e.g., render a styled UI component)?

2. Quick Summary of the Technology

- e.g.: “TailwindCSS is a utility-first CSS framework…”
- What is it?
- Where is it used?
- One real-world example.

3. System Requirements

- OS: Linux/Mac/Windows
- Tools/Editors required (e.g., VS Code, Node, Java JDK)
- Any packages (npm, pip, etc.)

4. Installation & Setup Instructions

- Step-by-step (with screenshots or terminal output if possible):
- **Example: Install Tailwind via npm*
- npm install -D tailwindcss
- npx tailwindcss init

5. Minimal Working Example

- Describe what the example does.

- Paste code with inline comments.
- Provide expected output.

6. AI Prompt Journal

- For each prompt used, record:

Prompt used:

- Link to the curriculum for the prompt
- "Give me a step-by-step guide to initialize TailwindCSS in a React app"
- AI's response summary: Optional
- Brief part of the response that addresses the problem
- "The AI helped me scaffold the setup and fix an import error..."
- Your evaluation of its helpfulness. Optional

7. Common Issues & Fixes

- What didn't work initially?
- Errors and how you resolved them.
- Links to StackOverflow, forums, etc.

8. References

- Official docs
- Video links
- Helpful blog posts

1. OBJECTIVE

As a data analyst, I often work with large datasets from different systems. I chose APIs to learn how to:

1. Fetch real-time data
2. Automate data collection
3. Improve my data analysis workflows

Goal

1. To build a simple Python program that:
2. Fetches live data from an API (currency exchange rates)
3. Extracts useful information
4. Displays it in a readable format

2. Quick Summary of the Technology What is an API?

An API (Application Programming Interface) allows applications to communicate with each other by sending requests and receiving responses.

Where is it used?

- Mobile apps (e.g., weather apps)
- Financial systems (stock prices, exchange rates)
- Web applications (login systems, maps)

Real-world Example

A currency exchange app uses an API to get live exchange rates between USD and KES.

3. System Requirements

- Operating System: Windows
- Tools Required: Python (3.14)
- Code editor (Jupyter notebook)
- Python Packages: matplotlib.pyplot, requests,pandas

```
pip install requests pandas
```

```
Requirement already satisfied: requests in c:\users\cynthia.dalmas\appdata\lo

Requirement already satisfied: pandas in c:\users\cynthia.dalmas\appdata\loca
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\cynthia.d
Requirement already satisfied: idna<4,>=2.5 in c:\users\cynthia.dalmas\appdat
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\cynthia.dalmas\
Requirement already satisfied: certifi>=2017.4.17 in c:\users\cynthia.dalmas\
Requirement already satisfied: numpy>=1.26.0 in c:\users\cynthia.dalmas\appda
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\cynthia.dal
Requirement already satisfied: pytz>=2020.1 in c:\users\cynthia.dalmas\appdat
Requirement already satisfied: tzdata>=2022.7 in c:\users\cynthia.dalmas\appd
Requirement already satisfied: six>=1.5 in c:\users\cynthia.dalmas\appdata\lo
```

4. Installation & Setup Instructions

- Install Python and Install required packages

- I will be usings JUPYTER NOTEBOOK

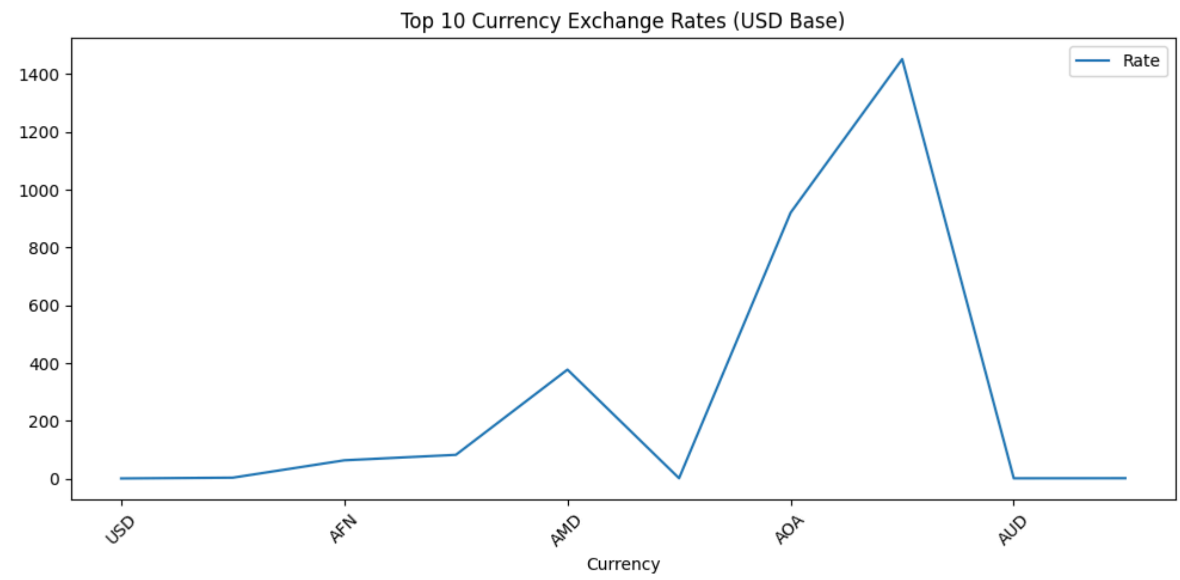
5. Minimal Working Example What the example does

- Connects to a currency exchange API
- Fetches exchange rates
- Extracts the USD to KES rate

```
#Packages needed
import pandas as pd
import matplotlib.pyplot as plt
import requests

url = "https://api.exchangerate-api.com/v4/latest/USD"
data = requests.get(url).json()
rates = data['rates']

df = pd.DataFrame(list(rates.items()), columns=['Currency','Rate'])


# Showing first few rows
print(df.head(10))
```

```
  Currency     Rate
0      USD     1.00
1      AED     3.67
2      AFN    63.79
3      ALL    82.76
4      AMD   377.66
5      ANG     1.79
6      AOA   920.84
7      ARS  1452.25
8      AUD     1.44
9      AWG     1.79
```

```
# Plot sample
df.head(10).plot(x='Currency', y='Rate',figsize=(10, 5))
plt.title("Top 10 Currency Exchange Rates (USD Base)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```




```
# Extracting Kenya Shilling (KES) rate
kes_rate = data['rates']['KES']
#tz_rate =data['rates']['TZS]

# Printing the result
print(f"1 USD = {kes_rate} KES")
```

```
1 USD = 129.5 KES
```

6. AI Prompt Journal

Prompt 1

Prompt used:

"Explain APIs in simple terms for a beginner"

AI Response Summary: Explained APIs as a way for systems to communicate and exchange data.

Usefulness: Very helpful in understanding the concept.

Prompt 2

Prompt used:

"How do I fetch data from an API using Python?"

AI Response Summary: Provided Python code using the requests library.

Usefulness: Helped me write my first working API call.

Prompt 3

Prompt used:

"How do I extract specific values from JSON in Python?"

AI Response Summary: Explained dictionary access and nested keys.

Usefulness: Helped me extract the KES exchange rate.

Prompt 4

Prompt used:

"Convert API JSON data into a pandas DataFrame"

AI Response Summary: Showed how to structure JSON into tabular format.

Usefulness: Helped me prepare data for analysis.

7. Common Issues & Fixes

- Issue 1: KeyError

Problem: Tried to access a key that didn't exist

Fix: Printed the full JSON to understand structure

- Issue 2: API not responding

Problem: Incorrect API URL

Fix: Verified endpoint from documentation

- Issue 3: JSON confusion

Problem: Didn't understand nested data

Fix: Used AI to explain JSON step-by-step

8. References

- Official Python Documentation https://docs.python.org/release/3.14.3/library/index.html
- Requests Library Documentation https://pypi.org/project/requests/
- Exchange Rate API website
- YouTube tutorials on APIs https://www.youtube.com/watch?v=ukBCjzC_C6w

GitHub README

https://github.com/cynthiadalmas/AI-CAPSTONE-PROJECT

```
from wordcloud import WordCloud


text = """
refactoring debugging testing creativity functionality chatgpt gemini claude
prompts java python flask api genai
refactoring debugging testing python python api genai creativity forex curren
"""

#  word cloud
wordcloud = WordCloud(
    width=900,
    height=450,
    background_color='white',
    colormap='viridis'
).generate(text)

#plotting
plt.figure(figsize=(12, 12))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("AI Capstone Project Word Cloud")

plt.show()
```